1. Executive Summary

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1. Executive summary
- Train games=940 | Derby games=75
- Nested CV=5 outer folds x 3 inner folds (expanding time-aware)
- Selected model family: Regime-aware simplex stack (Ridge/Huber/HistGB/GBR/XGB/HistGB-bag) + q50 blend
- Selected Elo variant: elo_base_static
- Nested outer RMSE/MAE: 36.674 / 29.057
- Selected postprocess module/eta: expand_affine_global / 0.55
- Calibration=none | Scale=none | Regime stack=False
- Runtime safety: fast_mode=False | budget_triggered=none
```

1B. Runtime Bottleneck And Fixes

Runtime bottleneck and fixes
- Profile-first instrumentation added for hotspot phase totals, per-family _predict_family timing, model fit counts, and top slow events.
- Shared split-level matrix/weight cache keyed by train/val index signatures + feature signature + half-life to eliminate repeated _xy/time-weight prep.
- Bundled family prediction path prepares X/y/w once per split and fits Ridge/Huber/HistGB/GBR/XGB/HistGB-bag/q50 sequentially.
- HistGB bagging reduced to 2 models by default (fast mode uses 1), with deterministic seed schedule and budget-aware auto-reduction.
- Candidate scan uses conservative early pruning: coarse score on first inner split, then full evaluation on top candidates only.
- Budget enforcement: total time, scan time, and fit cap stop further scanning/training and continue with best-so-far.
- Budgets triggered: none
- Fast mode=False; optimization toggles enabled=True.
- Reproducibility: seed=23, OMP_NUM_THREADS=1, MKL_NUM_THREADS=1.
- Artifact validation: predictions rows=75 (expected 75), rankings rows=165 (expected 165), ranks unique 1..165=True.

1C. Fast Profiling Before vs After (timing / fits) (1/1)

kind	name	before_sec	after_sec	delta_sec
phase	_scan_and_select_core_candidates	5.230	5.524	0.294
phase	_core_scan_score	5.226	5.516	0.290
phase	_generate_inner_oof_and_outer_preds	5.540	3.973	-1.567
phase	_predict_family	10.500	9.416	-1.084
phase	_build_split_tables	0.447	0.334	-0.113
phase	_prepare_variant_outer_data	0.550	0.440	-0.110
fit_count	histgb	48.000	51.000	3.000
fit_count	histgb_bag	8.000	6.000	-2.000
fit_count	histgb_q50	8.000	6.000	-2.000
fit_count	huber	8.000	6.000	-2.000
fit_count	ridge	48.000	51.000	3.000

1D. Runtime Focus Phase Timing Summary (1/1)

phase	total_sec	calls
_predict_family	2644.287	2890
_generate_inner_oof_and_outer_preds	1443.913	30
_scan_and_select_core_candidates	1181.556	6
_core_scan_score	1181.505	576
_prepare_variant_outer_data	3.926	12
_build_split_tables	3.492	48

1E. _predict_family Aggregated Timing / Fit Counts (1/1)

family	predict_time_sec	predict_calls	fit_count
histgb	1302.520	1081	1081
gbr	555.210	121	121
histgb_bag	251.450	121	242
xgb	243.044	121	121
histgb_q50	203.940	121	121
huber	81.456	121	121
ridge	3.638	1081	1081
histgb_q80	1.448	1	1
histgb_q20	1.362	1	1
enet_f2	0.220	121	121

2-4. Changes / Leakage / Nested CV

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2. What changed vs previous pipeline
- Nested tuning for post-model layers and key hyperparameters (Ridge alpha, HistGB grid, recency half-life).
- Regime-specific simplex stacking + regime-specific calibration + regime/pool scale correction.
- Recency features: EMA multi-alpha, last3/5/8, trend, consistency ratio.
- Upgraded Elo variants: dynamic-K, team-specific regularized home effect, inactivity decay.
- HistGB seed bagging and quantile HistGB (median, plus q20/q80 for diagnostics).
- Shift diagnostics (KS + PSI + out-of-range fraction) and OOD-triggered mitigation.

3. Data constraints and leakage controls
- All features are pregame-only and built sequentially.
- Static rating models refit inside each train split; no outer-fold leakage in postprocessing.
- Derby predictions are neutral-site (home effects zeroed for derby features).

4. Nested CV design (outer/inner schematic)
 outer_fold=1 train_n=125 val_n=193 train_end=2025-01-12 val=2025-01-23..2025-02-19
 outer_fold=2 train_n=318 val_n=205 train_end=2025-02-19 val=2025-03-03..2025-04-02
 outer_fold=3 train_n=523 val_n=151 train_end=2025-04-02 val=2025-04-14..2025-04-30
 outer_fold=4 train_n=674 val_n=102 train_end=2025-04-30 val=2025-05-13..2025-05-29
 outer_fold=5 train_n=776 val_n=164 train_end=2025-05-29 val=2025-05-30..2025-06-30
```

5. Outer-fold selections and metrics (1/1)

outer_candidate_id	dataset	feature_set	n_folds	lifecycle_weight	stacking_mode	calibration	robustness_penalty	mean_pr_auc	mean_roc_auc	mean_brier	mean_ece	mean_log_loss	mean_top_decile_lift	mean_positive_rate	mean_prediction_rate	mean_threshold	mean_n_oof	outer_n_oof	outer_pr_auc	outer_roc_auc	outer_brier	outer_ece	outer_log_loss	outer_top_decile_lift	outer_positive_rate	outer_prediction_rate	outer_threshold	outer_n_oof	delta_pr_auc	delta_roc_auc	delta_brier	delta_ece	delta_log_loss	delta_top_decile_lift	degraded	calibration_method	scale_pos_weight	n50_bootstrap	monotonic_cap
[illegible]	[illegible]	[illegible]	3	[illegible]	[illegible]	4-model	0.000	7	expand	raw	0.556	0.422	5.78	4.38	0.240	3.15	0.299	0.000	False	4.17	5.75	0.081	1.20	7.81	1.33	0.483	3.44	0.502	7.29	5.09	0.605	9.83	0.244	False	none	none	0.000	0.000	False
[illegible]	[illegible]	[illegible]	2	[illegible]	[illegible]	4-model	0.500	4	expand	raw	0.556	1.08	2.77	4.46	0.405	4.12	0.431	0.000	False	1.19	1.45	8.75	0.140	0.799	2.44	0.433	8.04	0.492	5.53	4.24	0.573	8.56	0.186	False	none	none	0.200	0.000	False
[illegible]	[illegible]	[illegible]	3	[illegible]	[illegible]	4-model	0.000	5	expand	raw	0.556	0.17	1.81	3.62	0.437	8.26	0.446	0.000	False	0.21	0.72	5.41	9.01	1.37	2.94	0.385	2.14	0.452	5.76	8.59	0.668	4.55	0.867	False	none	none	0.200	0.000	False
[illegible]	[illegible]	[illegible]	4	[illegible]	[illegible]	4-model	0.500	0	expand	raw	0.558	7.63	9.57	0.53	0.419	1.16	0.493	0.000	False	7.86	7.73	1.14	4.59	3.90	6.80	0.537	1.86	0.652	8.93	0.49	0.715	0.99	0.143	False	none	none	0.200	0.000	False
[illegible]	[illegible]	[illegible]	5	[illegible]	[illegible]	4-model	2.000	0	expand	raw	0.558	7.90	9.35	1.32	0.423	0.32	0.447	0.000	False	8.16	8.11	7.97	0.321	1.17	3.48	0.362	9.44	0.592	4.89	7.13	0.528	8.73	0.753	False	none	none	0.200	0.000	False

5. Outer-fold RMSE/MAE table (1/1)

outer_fold	rmse	mae	bias	pred_std	actual_std	resid_std	tail_rmse	xtail_rmse	tail_bias	tail_dispersion_rat	corr
1	40.080	31.204	-7.813	27.295	45.098	39.311	61.330	75.894	-13.443	0.483	0.501
2	38.753	30.143	-0.799	25.538	44.241	38.745	62.442	71.133	-18.042	0.433	0.490
3	35.416	29.019	1.374	25.760	38.591	35.389	52.947	56.895	-2.148	0.385	0.453
4	31.142	24.593	3.901	28.935	40.496	30.897	46.803	50.704	-1.863	0.537	0.650
5	37.979	30.326	1.170	24.898	47.133	37.961	63.485	70.295	-9.441	0.362	0.597

6. Feature additions and recency ablations (core scan) (1/1)

feature_profile	half_life_days	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
full_recency		43.331	42.695	53.676	480

6. Core scan results (top rows) (1/4)

candidate_ke	lo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber	scan_ridge	scan_stagep	ned_std_sca	scan_splits_u	scan_stage	outer_fold
_recency|hle	ele_base_sta	full_recenc		4.000	6	37.977	39.743	31.294					41.914	34.105	-28.960	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		4.000	6	37.977	39.743	31.294					41.914	34.105	-28.960	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		2.000	6	38.070	39.743	31.294					42.183	33.969	-28.960	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		2.000	6	38.070	39.743	31.294					42.183	33.969	-28.960	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		1.000	6	38.147	39.743	31.294					42.401	33.940	-28.960	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		1.000	6	38.147	39.743	31.294					42.401	33.940	-28.960	1	coarse	1
recency|hle	ele_base_sta	full_recenc		0.500	6	38.195	39.743	31.294					42.538	34.031	-28.960	1	coarse	1
recency|hle	ele_dynamic	full_recenc		0.500	6	38.195	39.743	31.294					42.538	34.031	-28.960	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		4.000	7	38.336	40.389	32.228					41.914	34.105	-29.043	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		4.000	7	38.336	40.389	32.228					41.914	34.105	-29.043	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		4.000	3	38.419	40.520	32.612					41.914	34.105	-29.249	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		4.000	3	38.419	40.520	32.612					41.914	34.105	-29.249	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		4.000	0	38.422	37.896	29.990					47.649	36.296	-15.553	3	full	4
_recency|hle	ele_dynamic	full_recenc		4.000	0	38.422	37.896	29.990					47.649	36.296	-15.553	3	full	4
_recency|hle	ele_base_sta	full_recenc		2.000	7	38.430	40.389	32.228					42.183	33.969	-29.043	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		2.000	7	38.430	40.389	32.228					42.183	33.969	-29.043	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		4.000	2	38.439	40.627	32.984					41.914	34.105	-28.480	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		4.000	2	38.439	40.627	32.984					41.914	34.105	-28.480	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		1.000	7	38.507	40.389	32.228					42.401	33.940	-29.043	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		1.000	7	38.507	40.389	32.228					42.401	33.940	-29.043	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		2.000	3	38.513	40.520	32.612					42.183	33.969	-29.249	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		2.000	3	38.513	40.520	32.612					42.183	33.969	-29.249	1	coarse	1
_recency|hle	ele_base_sta	full_recenc		2.000	2	38.533	40.627	32.984					42.183	33.969	-28.480	1	coarse	1
_recency|hle	ele_dynamic	full_recenc		2.000	2	38.533	40.627	32.984					42.183	33.969	-28.480	1	coarse	1

6. Core scan results (top rows) (2/4)

candidate_ke	lo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb	scan_gbr_sc	scan_gbr_sc	scan_huber	scan_huber	scan_ridge	scan_stagep	ned_std_sca	scan_splits_u	scan_stage	outer_fold
recency|hle	lo_base_sta	full_recenc		0.500	7	38.554	40.389	32.228					42.538	34.031	-29.043	1	coarse	1
recency|hle	lo_dynamic	full_recenc		0.500	7	38.554	40.389	32.228					42.538	34.031	-29.043	1	coarse	1
recency|hle	lo_base_sta	full_recenc		1.000	3	38.589	40.520	32.612					42.401	33.940	-29.249	1	coarse	1
recency|hle	lo_dynamic	full_recenc		1.000	3	38.589	40.520	32.612					42.401	33.940	-29.249	1	coarse	1
recency|hle	lo_base_sta	full_recenc		4.000	4	38.602	38.262	30.152					47.649	36.296	-15.217	3	full	4
recency|hle	lo_dynamic	full_recenc		4.000	4	38.602	38.262	30.152					47.649	36.296	-15.217	3	full	4
recency|hle	lo_base_sta	full_recenc		1.000	2	38.609	40.627	32.984					42.401	33.940	-28.480	1	coarse	1
recency|hle	lo_dynamic	full_recenc		1.000	2	38.609	40.627	32.984					42.401	33.940	-28.480	1	coarse	1
recency|hle	lo_base_sta	full_recenc		4.000	5	38.635	40.952	32.368					41.914	34.105	-28.831	1	coarse	1
recency|hle	lo_dynamic	full_recenc		4.000	5	38.635	40.952	32.368					41.914	34.105	-28.831	1	coarse	1
recency|hle	lo_base_sta	full_recenc		0.500	3	38.637	40.520	32.612					42.538	34.031	-29.249	1	coarse	1
recency|hle	lo_dynamic	full_recenc		0.500	3	38.637	40.520	32.612					42.538	34.031	-29.249	1	coarse	1
recency|hle	lo_base_sta	full_recenc		0.500	2	38.657	40.627	32.984					42.538	34.031	-28.480	1	coarse	1
recency|hle	lo_dynamic	full_recenc		0.500	2	38.657	40.627	32.984					42.538	34.031	-28.480	1	coarse	1
recency|hle	lo_base_sta	full_recenc		0.500	0	38.671	37.896	29.990					48.359	36.783	-15.553	3	full	4
recency|hle	lo_dynamic	full_recenc		0.500	0	38.671	37.896	29.990					48.359	36.783	-15.553	3	full	4
recency|hle	lo_base_sta	full_recenc		1.000	0	38.685	37.896	29.990					48.398	36.986	-15.553	3	full	4
recency|hle	lo_dynamic	full_recenc		1.000	0	38.685	37.896	29.990					48.398	36.986	-15.553	3	full	4
recency|hle	lo_base_sta	full_recenc		2.000	5	38.729	40.952	32.368					42.183	33.969	-28.831	1	coarse	1
recency|hle	lo_dynamic	full_recenc		2.000	5	38.729	40.952	32.368					42.183	33.969	-28.831	1	coarse	1
recency|hle	lo_base_sta	full_recenc		4.000	2	38.733	38.627	30.475					47.649	36.296	-14.353	3	full	4
recency|hle	lo_dynamic	full_recenc		4.000	2	38.733	38.627	30.475					47.649	36.296	-14.353	3	full	4
recency|hle	lo_base_sta	full_recenc		1.000	5	38.806	40.952	32.368					42.401	33.940	-28.831	1	coarse	1
recency|hle	lo_dynamic	full_recenc		1.000	5	38.806	40.952	32.368					42.401	33.940	-28.831	1	coarse	1

6. Core scan results (top rows) (3/4)

candidate_ke	lo_variat	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb	scan_gbr_is	scan_gbr_sc	scan_huber	scan_huber	scan_ridge	scan_stagep	ned_std_sca	scan_splits_u	scan_stage	outer_fold
recency|hle	ello_base_st	full_recenc		0.500	4	38.850	38.262	30.152					48.359	36.783	-15.217	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	4	38.850	38.262	30.152					48.359	36.783	-15.217	3	full	4
_recency|hle	ello_base_st	full_recenc		4.000	3	38.851	38.975	30.829					47.649	36.296	-13.364	3	full	4
_recency|hle	ello_dynamic	full_recenc		4.000	3	38.851	38.975	30.829					47.649	36.296	-13.364	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	5	38.853	40.952	32.368					42.538	34.031	-28.831	1	coarse	1
recency|hle	ello_dynamic	full_recenc		0.500	5	38.853	40.952	32.368					42.538	34.031	-28.831	1	coarse	1
_recency|hle	ello_base_st	full_recenc		4.000	7	38.893	39.055	30.853					47.649	36.296	-13.217	3	full	4
_recency|hle	ello_dynamic	full_recenc		4.000	7	38.893	39.055	30.853					47.649	36.296	-13.217	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	1	38.955	38.599	30.389					48.359	36.783	-14.005	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	1	38.955	38.599	30.389					48.359	36.783	-14.005	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	2	38.981	38.627	30.475					48.359	36.783	-14.353	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	2	38.981	38.627	30.475					48.359	36.783	-14.353	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	3	39.099	38.975	30.829					48.359	36.783	-13.364	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	3	39.099	38.975	30.829					48.359	36.783	-13.364	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	5	39.108	39.033	30.778					48.359	36.783	-12.506	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	5	39.108	39.033	30.778					48.359	36.783	-12.506	3	full	4
_recency|hle	ello_base_st	full_recenc		1.000	3	39.113	38.975	30.829					48.398	36.986	-13.364	3	full	4
_recency|hle	ello_dynamic	full_recenc		1.000	3	39.113	38.975	30.829					48.398	36.986	-13.364	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	7	39.142	39.055	30.853					48.359	36.783	-13.217	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	7	39.142	39.055	30.853					48.359	36.783	-13.217	3	full	4
_recency|hle	ello_base_st	full_recenc		1.000	7	39.156	39.055	30.853					48.398	36.986	-13.217	3	full	4
_recency|hle	ello_dynamic	full_recenc		1.000	7	39.156	39.055	30.853					48.398	36.986	-13.217	3	full	4
recency|hle	ello_base_st	full_recenc		0.500	6	39.281	39.448	31.012					48.359	36.783	-11.675	3	full	4
recency|hle	ello_dynamic	full_recenc		0.500	6	39.281	39.448	31.012					48.359	36.783	-11.675	3	full	4

6. Core scan results (top rows) (4/4)

candidate_ke	elo_variant	feature_pro	half_life_da	ridge_alph	histgb_idx	scan_score	scan_histgb	scan_histgb_	scan_gbr_rs	scan_gbr_sc	scan_huber_	scan_huber_	scan_ridge_s	scan_ridge_	scan_stagepred_std_sca	scan_splits_u	scan_stage	outer_fold
_recency|hle	elo_base_sta	full_recenc		4.000	3	40.048	40.179	31.835					48.869	35.878	-15.919	3	full	3
_recency|hle	elo_dynamic	full_recenc		4.000	3	40.048	40.179	31.835					48.869	35.878	-15.919	3	full	3
_recency|hle	elo_base_sta	full_recenc		4.000	7	40.230	40.510	31.997					48.869	35.878	-15.869	3	full	3
_recency|hle	elo_dynamic	full_recenc		4.000	7	40.230	40.510	31.997					48.869	35.878	-15.869	3	full	3
_recency|hle	elo_base_sta	full_recenc		4.000	5	40.294	40.812	31.974					48.869	35.878	-14.208	3	full	3
_recency|hle	elo_dynamic	full_recenc		4.000	5	40.294	40.812	31.974					48.869	35.878	-14.208	3	full	3
recency|hle	elo_base_sta	full_recenc		0.500	7	40.304	41.012	33.071					47.435	37.053	-18.579	3	full	2
recency|hle	elo_dynamic	full_recenc		0.500	7	40.304	41.012	33.071					47.435	37.053	-18.579	3	full	2

7. Elo variant comparisons (1/1)

elo_variant	scan_score_mean	scan_histgb_rmse_mean	scan_ridge_rmse_mean	n_configs
elo_base_static	43.331	42.695	53.676	240
elo_dynamic_k	43.331	42.695	53.676	240

7. HistGB hyperparameter sweep summary (1/1)

histgb_idx	scan_score_mean	scan_histgb_rmse_mean	n	params
4	42.749	41.678	62	{'learning_rate': 0.06, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 12
3	42.953	41.865	64	{'learning_rate': 0.0256, 'max_depth': 5, 'max_leaf_nodes': 63, 'min_samples_leaf': 5
5	42.852	41.984	58	{'learning_rate': 0.07, 'max_depth': 5, 'max_leaf_nodes': 63, 'min_samples_leaf': 4,
7	42.891	42.107	68	{'learning_rate': 0.08, 'max_depth': 4, 'max_leaf_nodes': 127, 'min_samples_leaf': 5
2	43.100	42.132	60	{'learning_rate': 0.03, 'max_depth': 4, 'max_leaf_nodes': 63, 'min_samples_leaf': 6,
6	43.442	42.869	64	{'learning_rate': 0.05, 'max_depth': 6, 'max_leaf_nodes': 127, 'min_samples_leaf': 3
1	44.055	43.762	48	{'learning_rate': 0.05, 'max_depth': 4, 'max_leaf_nodes': 31, 'min_samples_leaf': 8,
0	44.937	45.709	56	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10

8. Regime-specific stacking / calibration results (1/1)

module_name	use_regime_sta	calibration_mod	scale_mode	eta	outer_rmse_mo	outer_mae_outer	outer_tail_dispersion	outer_tail_bias_m	outer_corr_mean	outer_dispersion_	regime_ratio	gain_vs_poole	n
xpand_affine_gl	False	none	none	0.550	36.639	29.049	57.505	0.435	-9.053	0.537	0.614	-0.101	25

9. Dispersion and scale correction analysis (1/1)

outer_fold	stage	pred_std	actual_std	rmse
1	raw_mean	27.295	45.098	39.947
1	calibrated	27.295	45.098	39.947
1	scaled	27.295	45.098	39.947
1	final	27.295	45.098	40.080
2	raw_mean	25.754	44.241	38.879
2	calibrated	25.538	44.241	38.878
2	scaled	25.538	44.241	38.878
2	final	25.538	44.241	38.753
3	raw_mean	26.116	38.591	35.495
3	calibrated	25.760	38.591	35.396
3	scaled	25.760	38.591	35.396
3	final	25.760	38.591	35.416
4	raw_mean	29.436	40.496	30.806
4	calibrated	28.935	40.496	30.908
4	scaled	28.935	40.496	30.908
4	final	28.935	40.496	31.142
5	raw_mean	24.972	47.133	37.915
5	calibrated	24.898	47.133	37.979
5	scaled	24.898	47.133	37.979
5	final	24.898	47.133	37.979

10. Shift diagnostics: Train recent vs Derby (1/1)

feature	train_recent_mean	derby_mean	train_recent_std	derby_std	ks_stat	ks_pvalue	psi	derby_outside_train_1_9	ood_flag
games_played_min	9.122	11.067	1.095	0.660	0.730	0.000	8.252	0.227	True
elo_diff_pre	58.187	-0.325	64.506	58.625	0.409	0.000	2.070	0.107	True
same_conf_flag	0.697	0.013	0.459	0.115	0.684	0.000	0.000	0.000	True
massey_diff	5.872	2.195	19.548	15.187	0.166	0.075	0.192	0.000	False
consistency_ratio_di	0.059	-0.008	0.657	0.577	0.113	0.427	0.167	0.013	False
_margin_last5_vs_sea	-1.434	1.554	16.927	19.606	0.105	0.514	0.133	0.067	False
volatility_sum	71.164	73.217	11.169	9.954	0.139	0.196	0.132	0.027	False
offdef_net_diff	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
offdef_margin_neutr	1.077	-0.266	22.991	17.140	0.120	0.346	0.115	0.000	False
elo_neutral_diff_pre	3.187	-0.325	64.506	58.625	0.099	0.591	0.053	0.013	False

10. Shift Mitigation Notes

```
10. Shift mitigation rules
- Detected 3 OOD-like feature shifts (KS/PSI/out-of-range checks).
- Applied softer scaling (blend toward 1.0) and wider winsor bounds under OOD risk.
- No anti-compression guard applied (derby std 25.90, ref 25.30).
```

8. Regime weights (outer_fold_1_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	103	0.136	0.432	0.000	0.000	0.432	0.000

8. Regime weights (outer_fold_2_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	284	0.147	0.261	0.000	0.038	0.261	0.293

8. Regime weights (outer_fold_3_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	448	0.028	0.083	0.006	0.512	0.083	0.289

8. Regime weights (outer_fold_4_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	549	0.095	0.000	0.672	0.000	0.000	0.233

8. Regime weights (outer_fold_5_selected) (1/1)

level	regime	n	w_pred_ridge	w_pred_histgb	w_pred_gbr	w_pred_xgb	w_pred_histgb_bag	w_pred_enet_f2
pooled	ALL	589	0.000	0.172	0.000	0.000	0.172	0.656

8. Calibration params (outer_fold_1_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	103	0.000	1.000

8. Calibration params (outer_fold_2_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	284	0.000	1.000

8. Calibration params (outer_fold_3_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	448	0.000	1.000

8. Calibration params (outer_fold_4_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	549	0.000	1.000

8. Calibration params (outer_fold_5_selected) (1/1)

level	regime	n	intercept	slope
pooled	ALL	589	0.000	1.000

9. Scale params (outer_fold_1_selected) (1/1)

level	regime	n	scale
pooled	ALL	103	0.950

9. Scale params (outer_fold_2_selected) (1/1)

level	regime	n	scale
pooled	ALL	284	0.900

9. Scale params (outer_fold_3_selected) (1/1)

level	regime	n	scale
pooled	ALL	448	0.800

9. Scale params (outer_fold_4_selected) (1/1)

level	regime	n	scale
pooled	ALL	549	0.900

9. Scale params (outer_fold_5_selected) (1/1)

level	regime	n	scale
pooled	ALL	589	0.800

11. Residual Diagnostics (Summary)

```
11. Residual diagnostics
- OOF final RMSE=37.446, MAE=29.528, Bias=-1.073
- Residual std=37.430, skew=0.180, kurtosis=0.197
- Mean-stack RMSE=37.411
- Quantile median (q50) RMSE=37.862
- Mean+median blend RMSE=37.417
- Calibrated RMSE=37.417
- Tail error threshold (top 10%)=61.618
```

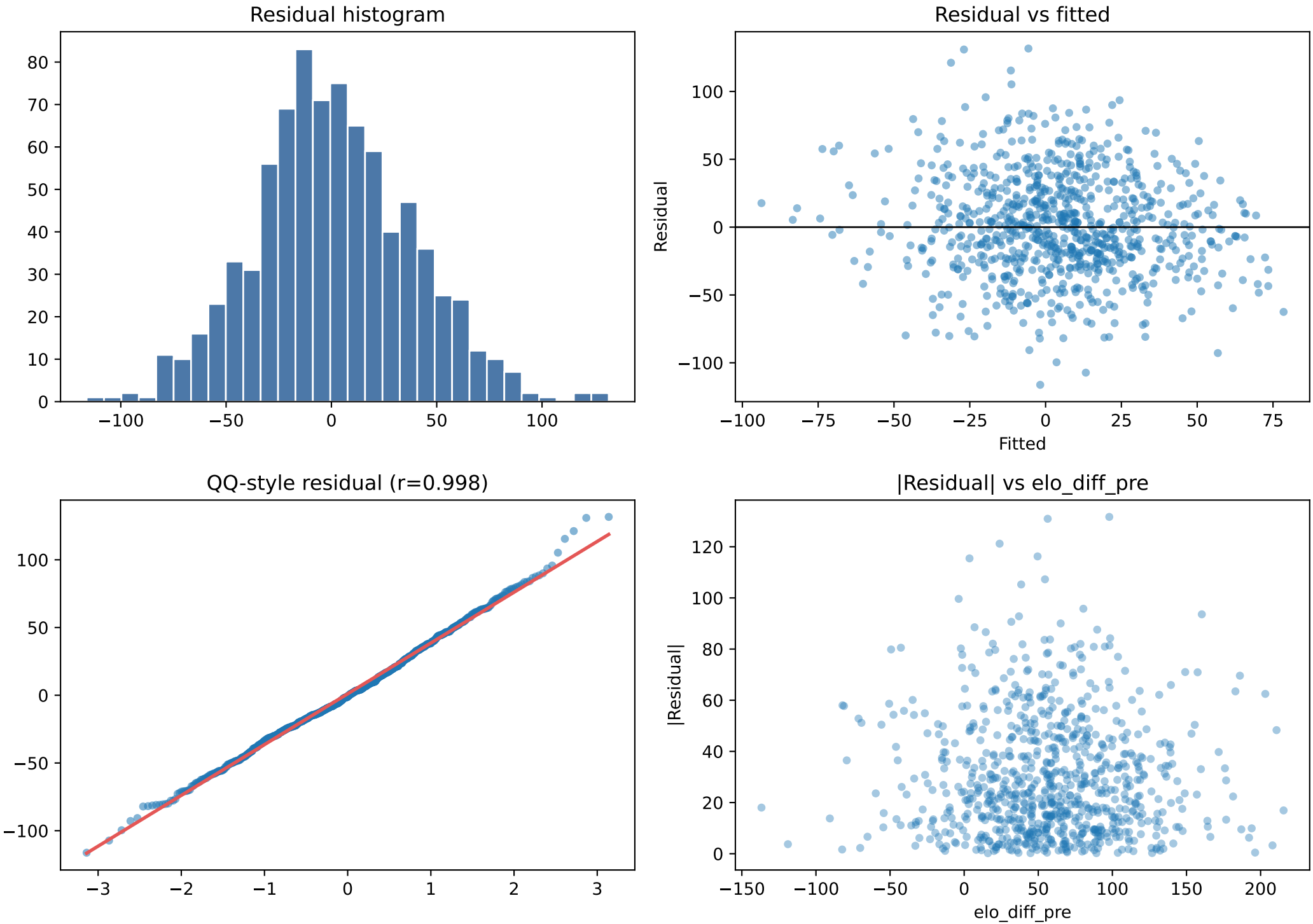
11. Tail error diagnostics (top 10% abs-error) (1/2)

| Date | ted_diff | hassel | effect | volatil | volgain | vsgain | phase_p | lagrec | lagrec | lagrec

11. Tail error diagnostics (top 10% abs-error) (2/2)

Date	ret_diff	massoff	eff_vol	volatility	volatility	gap	gap_pct	bar_range	bar_range_pct	bar_dir	bar_body_pct	bar_upper_wick	bar_lower_wick	bar_pos_in_range	bar_streak	row_idx	y_true	pred_prob	pred_eprob	pred_lprob	pred_hist	pred_qhist	pred_qbin	eff_n	raw_raw	raw_cal	pad_score	pad_frac	pad_idx	process	abs_error
-04-02	0	8.047	4.158	5.253	6.683	4.573	5.000	7.000	1.000	0.000	-0.743	-1.131	-0.032	513	-62.00	5.461	7.350	8.141	5.098	6.391	6.455	5.092	1.41	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.389e-0
-01-24	0	4.906	4.124	7.107	5.000	7.000	1.000	2.000	0.000	1.000	0.000	0.000	-1.690	159	-60.00	2.416	-1.743	3.592	4.703	6.112	2.562	4.703	19.46	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.965e-
-04-02	0	7.472	3.587	4.156	8.085	7.196	6.000	5.000	1.000	0.000	-1.533	-1.836	0.342	479	-48.00	2.803	8.908	3.133	8.363	9.409	0.872	8.363	33.52	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.892e-0
-03-18	0	4.599	5.417	-0.036	6.208	0.146	6.000	5.000	1.000	0.000	7.267	7.211	0.122	405	84.00	2.481	-3.511	3.186	-1.260	2.799	-5.301	-1.260	4.329	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.763e-0
-06-16	0	2.695	4.847	6.282	4.985	1.395	1.000	1.000	0.000	1.000	1.745	3.421	1.457	907	-104.00	5.957	9.787	4.799	5.585	4.283	3.585	5.585	28.15	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.535e-
-02-19	0	-2.217	-0.798	-8.000	0.625	9.708	3.000	4.000	1.000	0.000	0.000	0.000	0.717	267	-112.00	7.027	5.681	1.795	1.124	8.804	4.311	1.124	24.67	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.246e-
-03-18	0	2.055	2.626	-6.389	3.744	4.790	3.000	4.000	1.000	0.000	0.000	0.000	-0.118	437	68.00	1.823	0.754	-4.342	9.669	8.141	6.595	9.669	18.07	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.224e-0
-06-16	0	9.309	8.505	-2.008	4.463	4.372	0.000	0.000	1.000	0.000	5.400	6.567	-0.336	899	-126.00	3.633	7.505	6.835	1.524	8.615	7.785	1.524	46.93	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.849e-
-02-19	0	0.938	3.589	-6.250	6.766	6.173	3.000	4.000	1.000	0.000	0.000	0.000	1.143	280	36.00	4.800	4.162	0.354	8.949	6.935	5.764	8.949	20.80	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.678e-
-03-04	0	6.889	2.738	1.378	6.949	5.477	3.000	4.000	1.000	0.000	0.000	0.000	-1.635	382	70.00	5.991	6.163	9.134	5.644	7.859	4.924	5.644	74.02	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.662e-0
-05-16	0	6.707	8.934	4.691	7.808	2.675	7.000	9.000	0.000	1.000	3.530	3.740	-0.673	706	66.00	8.344	7.712	2.917	2.505	1.522	2.174	2.505	18.55	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.625e-
-04-14	0	5.756	8.391	1.332	6.694	4.187	6.000	6.000	0.000	1.000	8.267	8.197	0.678	531	44.00	7.847	9.124	0.409	9.888	8.494	4.313	9.888	26.37	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	1.196e-
-04-16	0	8.114	1.300	-0.333	1.988	6.254	6.000	6.000	0.000	1.000	0.667	0.666	0.324	542	-80.00	6.469	0.464	0.404	5.439	0.735	0.985	5.439	0.220	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	7.825e-
-05-30	0	-1.304	5.184	9.895	4.529	6.609	1.000	0.000	0.000	1.000	3.582	3.752	0.377	829	-14.00	6.897	7.834	8.035	2.834	9.374	3.724	2.834	47.10	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	7.773e-
-06-27	0	3.768	8.635	6.943	4.683	7.918	0.000	0.000	1.000	0.000	4.000	3.373	0.166	916	98.00	7.163	4.545	3.096	0.236	2.319	0.068	0.236	4.66	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	6.980e-
-02-05	0	1.541	2.175	-5.000	9.394	3.000	3.000	2.000	1.000	0.000	0.000	0.000	-3.571	196	-102.00	9.782	3.309	6.105	1.624	2.534	2.524	1.624	27.38	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	6.626e-
-04-15	0	9.619	3.078	-2.476	2.024	6.438	8.000	6.000	1.000	0.000	8.867	9.286	-0.097	534	64.00	1.431	9.907	1.354	4.021	5.246	8.214	4.021	16.03	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	6.583e-
-04-02	0	3.445	4.008	-0.800	0.452	7.586	5.000	5.000	1.000	0.000	0.000	0.000	-0.204	511	76.00	3.531	4.932	0.175	7.545	8.353	0.903	7.545	10.65	module=[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	cal=non	6.150e-0

11. Residual diagnostics plots



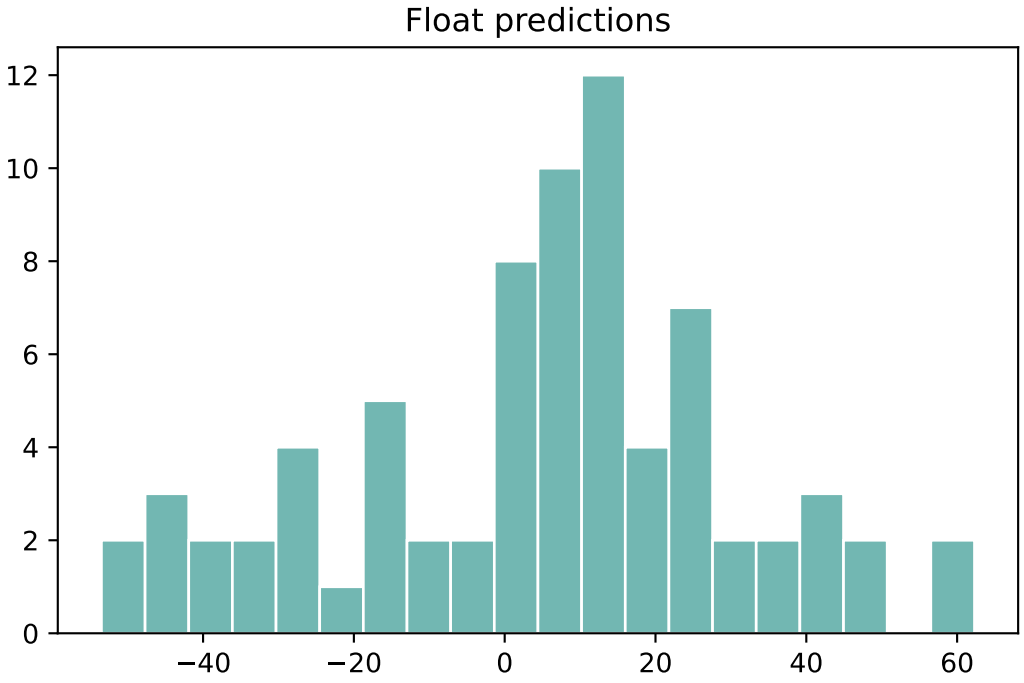
12-14. Final Decisions / Limitations / Next Steps

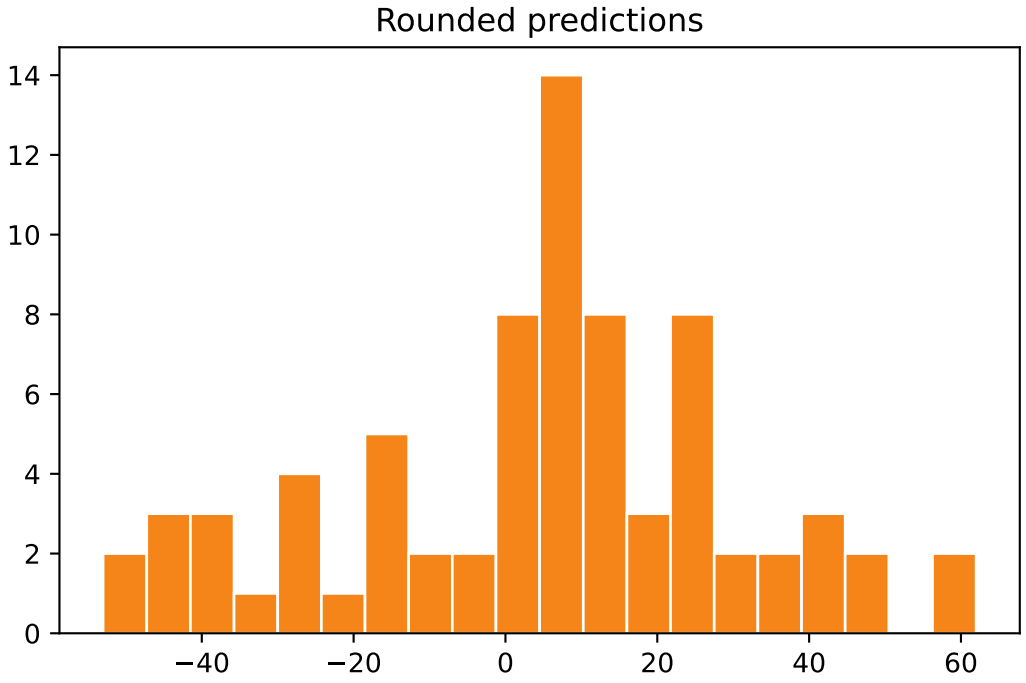
```
12. Final derby prediction distribution
- count=75, mean=4.360, std=25.895
- min=-53.478, max=62.303
- quantiles 1/5/25/50/75/95/99 = -50.211, -44.229, -11.776, 7.534, 21.491, 44.812, 61.342

13. Final decisions and rationale
- Final core candidate: elo_base_static|full_recency|hl=none|ridge=0.5|hgb=0
- Final postprocess candidate: module=expand_affine_global|stack=pool|cal=none|scale=none|q50=0.20|win=0.000|eta=0.55
- Recency profile=full_recency | half_life=None
- Shift mitigation applied=True
- Dispersion guard applied=False

14. Limitations + next steps
- Derby labels hidden: final selection relies on nested train-only proxies.
- No injuries/lineups/travel features available.
- Future work: monotone piecewise regime calibration and richer Bayesian uncertainty.
```

12. Final derby prediction distribution





Appendix A. First 10 rows of predictions.csv (1/1)

GameID	Date	Team1_Conf	Team1_ID	Team1	Team2_Conf	Team2_ID	Team2	Team1_WinMargin
1941	2025-07-04	Purple	68	Idaho	Yellow	119	Houston	5
1942	2025-07-04	White	105	Helena	Yellow	120	Iowa	21
1943	2025-07-04	Red	82	Jackson	Purple	67	Concord	5
1944	2025-07-04	White	108	Montpelier	Yellow	129	West Virginia	10
1945	2025-07-04	Yellow	121	Lansing	White	113	South Dakota	-13
1946	2025-07-04	Crimson	21	Lincoln	Yellow	118	Florida	23
1947	2025-07-04	Yellow	117	Columbia	Crimson	23	Nashville	6
1948	2025-07-04	Orange	63	New Jersey	Yellow	126	Pierre	28
1949	2025-07-04	Purple	77	Wilmington	Orange	58	Honolulu	6
1950	2025-07-04	Crimson	22	Michigan	Green	49	Natchez	16

Appendix B. Top 20 rankings by Rank (1/1)

TeamID	Team	Rank
36	Massachusetts	1
90	Big Sur	2
94	Kentucky	3
37	Minnesota	4
40	Pennsylvania	5
45	Boise City	6
2	Baton Rouge	7
153	Tokyo	8
26	Phoenix	9
92	Illinois	10
88	Port St. Lucie	11
137	Milan	12
46	Boston	13
54	Boise	14
47	Chicago	15
19	Georgia	16
13	Texas	17
85	New Hope	18
131	Belgrade	19
149	Seoul	20

Appendix C. Selected hyperparameters (1/1)

name	value
elo_variant	elo_base_static
feature_profile	full_recency
half_life_days	none
ridge_alpha	0.5
histgb_idx	0
histgb_params	{'learning_rate': 0.04, 'max_depth': 3, 'max_leaf_nodes': 31, 'min_samples_leaf': 10, 'l2_regularization': 0.3, 'ma
postprocess	module=expand_affine_global|stack=pool|cal=none|scale=none|q50=0.20|win=0.000|eta=0.55
postprocess_module	expand_affine_global
postprocess_eta	0.55
winsor_bounds	(-116.61, 113.305)
dispersion_guard_factor	1.0
seed	23
OMP_NUM_THREADS	1
MKL_NUM_THREADS	1

Appendix D. Final derby stage outputs (head) (1/1)

Date	elo_diff	passes	off_ref	volatility	volatile	save	pass_pct	tackle_pct	dribble_pct	consistency	row_index	pred_prob	pred_emp	pred_hum	pred_hist	pred_gr	pred_sq	pred_hist	pred_q5	pred_q2	stage8_s	stage_sum	stage_cat	stage_edge	stage_fin	spread_q
-07-04 0	-4.319	-0.364	-3.797	35.596	29.274	11.000	12.000	0.000	1.000	11.839	11.198	-0.229	0	25.220	2.229	56.095	0.745	0.940	5.082	0.745	7.543	25.321	15.504	1.656	2.834	2.834	2.834	4.695	40.827
-07-04 0	22.378	14.106	12.458	39.164	40.902	10.000	12.000	0.000	1.000	-20.806	-21.107	0.499	1	44.565	26.609	55.285	12.549	13.349	3.399	12.549	12.330	-6.491	33.944	20.809	19.113	19.113	19.113	20.974	40.435
-07-04 0	-61.146	-3.146	-7.864	59.498	27.316	11.000	13.000	0.000	1.000	9.849	10.915	-0.544	2	-36.969	4.342	-38.078	2.056	-3.434	12.134	2.056	7.674	-22.462	34.915	1.412	2.664	2.664	2.664	4.525	57.377
-07-04 0	27.445	2.010	-4.412	44.919	28.756	10.000	11.000	0.000	1.000	21.600	19.898	0.109	3	48.840	2.267	69.243	13.866	17.055	7.556	13.866	8.271	-17.366	30.093	8.413	8.385	8.385	8.385	10.246	47.453
-07-04 0	-20.059	10.874	-16.758	38.940	36.984	12.000	10.000	0.000	1.000	-5.000	-4.746	-0.546	4	-17.339	15.724	13.763	14.463	-13.483	19.633	14.463	16.303	-45.870	1.793	-14.833	15.126	15.126	15.126	13.264	47.662
-07-04 0	21.386	8.446	1.765	30.409	33.738	11.000	11.000	0.000	1.000	-21.524	-21.221	0.054	5	37.464	15.908	62.904	19.455	23.546	25.078	19.455	28.712	0.306	39.217	18.890	20.854	20.854	20.854	22.715	38.911
-07-04 0	35.060	1.343	2.118	40.055	39.435	11.000	11.000	0.000	1.000	-2.073	-3.274	-0.082	6	31.839	0.183	58.496	3.492	7.775	4.561	3.492	9.582	-16.821	16.022	3.129	4.420	4.420	4.420	6.281	32.843
-07-04 0	10.828	14.158	8.985	35.920	32.824	10.000	11.000	0.000	1.000	7.418	7.978	0.381	7	58.580	28.304	88.445	23.638	26.813	24.565	23.638	21.119	1.060	43.763	27.408	26.150	26.150	26.150	28.011	42.703
-07-04 0	-34.104	0.807	-0.588	41.956	41.415	11.000	11.000	0.000	1.000	-17.094	-17.566	-0.023	8	-12.615	8.782	5.562	-5.282	-0.792	-8.480	-5.282	4.776	-16.001	16.976	4.285	4.383	4.383	4.383	6.245	32.982
-07-04 0	66.423	13.143	21.640	28.163	37.988	10.000	11.000	0.000	1.000	-12.924	-14.346	1.020	9	-0.841	18.772	6.269	4.640	9.477	-3.804	4.640	13.395	-14.973	35.233	14.368	14.173	14.173	14.173	16.034	60.211
-07-04 0	35.309	6.094	1.544	41.287	34.893	10.000	11.000	0.000	1.000	-3.855	-3.785	0.003	10	34.197	9.222	52.848	11.220	12.582	7.877	11.220	7.052	-8.678	27.022	10.568	9.865	9.865	9.865	11.726	35.699
-07-04 0	-11.625	-4.250	-8.824	32.105	38.441	11.000	11.000	0.000	1.000	-9.964	-9.016	-0.303	11	18.586	-4.052	45.517	1.057	-1.166	-0.103	1.057	-0.217	12.801	19.692	-2.628	-2.146	-2.146	-2.146	-0.285	32.498
-07-04 0	-18.207	1.607	0.608	31.302	40.159	11.000	12.000	0.000	1.000	-0.827	-0.219	0.101	12	1.884	8.195	27.362	-9.289	-9.186	11.751	-9.289	6.222	-7.883	22.206	0.667	1.778	1.778	1.778	3.639	30.089
-07-04 0	-11.916	-2.432	-8.588	34.862	33.405	12.000	11.000	0.000	1.000	-12.904	-12.476	-0.039	13	-2.787	-0.546	10.085	-4.421	-2.076	-7.919	-4.421	3.598	-18.615	14.714	-1.393	-0.395	-0.395	-0.395	1.466	33.329
-07-04 0	42.745	7.018	9.059	33.643	31.326	11.000	11.000	0.000	1.000	1.200	-0.538	0.401	14	26.223	10.086	31.140	1.369	5.221	-3.155	1.369	-2.340	17.201	18.682	7.677	5.673	5.673	5.673	7.534	35.883
-07-04 0	26.821	0.733	-3.882	33.735	36.430	11.000	11.000	0.000	1.000	-19.606	-20.081	-0.155	15	18.733	0.196	35.648	6.675	-0.454	16.959	6.675	8.896	-14.913	31.328	0.479	2.162	2.162	2.162	4.023	46.239
-07-04 0	-10.571	5.344	-0.026	28.877	37.659	12.000	11.000	0.000	1.000	11.533	12.246	0.065	16	17.925	14.478	38.843	1.614	2.982	4.331	1.614	10.609	-14.379	18.650	9.396	9.639	9.639	9.639	11.500	33.030
-07-04 0	19.333	-0.458	-2.706	58.962	38.414	11.000	12.000	0.000	1.000	41.327	40.959	-0.193	17	16.129	-1.446	31.559	29.904	-26.803	-27.942	-29.904	10.085	-43.399	14.516	12.663	-12.148	-12.148	-12.148	10.285	57.909
-07-04 0	31.221	6.648	-6.941	33.206	35.506	11.000	11.000	0.000	1.000	2.327	3.044	-0.150	18	5.473	19.738	26.913	25.218	28.693	27.519	25.218	28.779	-6.341	36.895	23.338	24.426	24.426	24.426	26.287	30.553
-07-04 0	7.150	-3.761	-6.333	44.267	37.696	12.000	12.000	0.000	1.000	35.100	33.916	-0.273	19	22.865	-5.901	41.330	27.124	-28.494	-35.883	-27.124	18.253	-44.191	2.433	-15.566	-16.105	-16.105	-16.105	14.244	46.630